SMART INDIA HACKATHON 2026

RASAD (रसद)

Decide before the road shuts

- **Problem Statement ID –** SIH26251
- **Problem Statement Title-** Indian Army: Predictive Logistics & Forward Supply Chain
- **Theme-** Transportation & Logistics
- **PS Category-** Software
- **Team ID-** 163616
- **Team Name-** Code ZCrew



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RASAD: DECIDE BEFORE THE ROAD SHUTS

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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

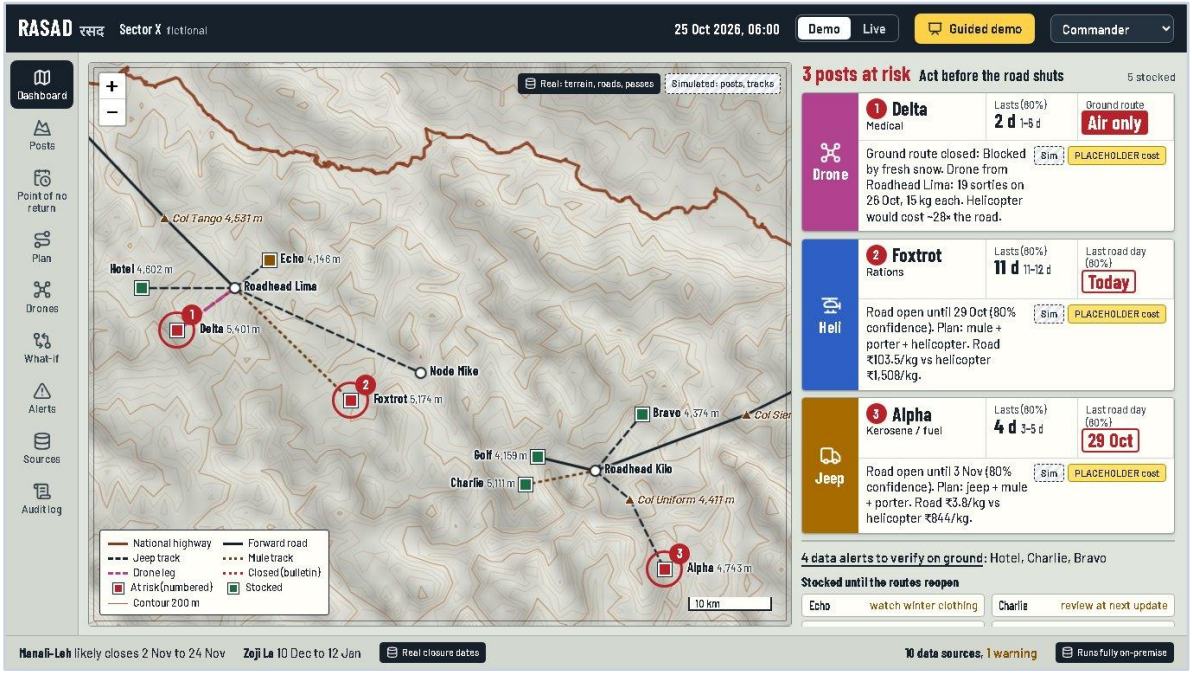
Forecast need → **Last safe road day** → **Mixed-fleet plan** → **Commander approves**

How it addresses the problem

Joins **10 separate data streams** into one decision picture

Innovation and uniqueness of the solution

- **Closure probability window**
- **Drones as a real mode**
- **Drone basing advisor**
- **Offline, on-premise**



Working prototype · military data simulated · terrain, highways, weather real

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TECHNICAL APPROACH

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Technologies to be used (e.g. programming languages, frameworks, hardware)

AI / ML
LightGBM quantile forecast (P10 / P50 / P90)

Optimisation
Google OR-Tools CP-SAT solver

GIS
Leaflet map · Copernicus DEM · OpenStreetMap

Backend
Python · FastAPI · pandas · SQLite

Frontend
React + TypeScript web app

Hardware
One on-prem laptop or server · IoT fuel sensors

Methodology and process for implementation (Flow Charts/Images/ working prototype)

10 data streams
Real: weather since 1995, 16-day forecast, terrain (DEM), highways (OSM)
Simulated: CICG-like stock & issues, IoT fuel sensors, BRO roads, IAF airlift

→

Adapters
1 config file per stream
Unknown codes refused
Health checks

→

One database
SQLite on the unit's own machine

→

Engines
Forecast (LightGBM) → **Point of No Return** (31 real winters) → **Mixed-fleet plan** (CP-SAT, 7 days)
+ drone advisor · what-if · data alerts

→

Commander
9 screens
Approve / Modify / Reject
Audit log



Working prototype

Every plan line explains itself: route, terrain clearance, sorties and cost vs helicopter, then Approve / Modify / Reject.

Costs are PLACEHOLDER.

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

- ✓ **Built and running:** 9 screens, one-click start, guided demo
- ✓ **Works offline** on one laptop; no data leaves it
- ✓ **Free, open-source stack:** no licence or cloud cost
- ✓ **Plugs into existing systems:** reads CICG, BRO and IAF exports without changing them
- ✓ **Stress-tested:** troops ×1.5 → posts at risk 3 → 6; pass closes 10 days early → helicopter sorties 30 → 49

Limitation → Mitigation *our honest gaps, and how we handle them*

Limitation (potential challenges and risks)	Mitigation (strategies for overcoming these challenges)
Military data is classified, so our stock and use data are simulated	Adapters read real CICG, BRO and IAF exports through one config file each; simulated data is labelled on every screen
Forecast accuracy is measured only on simulated data, built from the same weather inputs	The model is used only where it beats a 28-day average on held-out days; real accuracy is measured once CICG data is connected
Forward-track closure dates use PLACEHOLDER scaling of the pass rule	Pass rules are fitted on real closure dates; the unit's own track records replace the placeholders
Only 4–5 real closures per pass (error ±13 / ±18 days)	Plan at 80% confidence, not the middle date; add BRO's records as they come
The demo scenario is designed, and all costs are PLACEHOLDER	The same engine was re-run with changed inputs and responded correctly; real costs go into one parameter file
Large sectors may hit the solver's 20-second limit; roles are demo-only	The screen then says "best plan found within time limit"; Army SSO/PKI supplies roles in deployment

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IMPACT AND BENEFITS

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Potential impact on the target audience

For **commanders, logistics staff and post quartermasters**

29 Oct

Alpha's last safe road day, at 80% confidence, known before the pass closes

simulated · PLACEHOLDER track rules

30 → 0

Foxtrot's helicopter sorties this week, with 20 drones at the advised base

simulated · PLACEHOLDER costs

Benefits of the solution (social, economic, environmental, etc.)

Decide before passes close
Last safe road day for every post and supply

Less risk to troops
Fewer risky helicopter sorties and porter / mule trips

Ready for 2,715 drones
Advises where to base them, by altitude band

Full accountability
Every approve, modify or reject is logged

Plugs into existing systems
Reads CICG and other exports; replaces nothing

Scales to other sectors
Any high-altitude sector: new data, same engine

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RESEARCH AND REFERENCES

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Details / Links of the reference and research work

Problem and policy

MoD RFI: 2,715 logistic drones, 3 altitude bands (Aug 2026)
idrw.org · Outlook

Army RFP: 363 drones, 15 kg at high altitude, 10 km range
The Tribune, 17 Oct 2022

Joint Primer for Integrated Logistics, HQ IDS (Aug 2025)
PIB via GlobalSecurity

Winter stocking: ~2 lakh t for ~6 months; cost quotes
The Week, 13 Aug 2020

Mule units retained to ~2032
SP's Land Forces, Nov 2025

Q-FORCE logistics app (May 2026)
AIR News, 22 May 2026

Full list with confidence levels: docs/FACTS.md

Real closure dates

Zoji La closures and reopenings, 2020–25
Greater Kashmir · The Tribune · ThePrint · AIR News

Manali–Leh closures and openings, 2021–26
AIR News 2025 · AIR News 2024 · himalayan-routes

Open data used (real, cached offline)

Weather since 1995 and 16-day forecast
Open-Meteo ERA5 (CC BY 4.0)

Terrain heights and national highways
Copernicus DEM GLO-90 · OpenStreetMap (ODbL)

Logistics drone specs: Sabal 20
The Defense Post, Dec 2024

All data is free and public; no account or key needed

What we verified

- ✓ **Real data:** 31 winters of weather, terrain, highways and closure dates
- ✓ **Prior work:** 3 public prototypes for this problem studied; none plans drones or gives a closure probability
- ✓ **Self-check:** outputs change correctly when inputs change. Military data simulated; costs PLACEHOLDER